

Program overview

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Year 2022/2023
Organization Technology, Policy and Management
Education Minors WM

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Organization	Technology, Policy and Management
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Minors WM 2022

Year	2022/2023
Organization	Technology, Policy and Management
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WM-Mi-099-22	
Minor Coordinator	K.L.L. van Nunen
Administration by the Faculty of	TBM
Minor Title	Minor Security, Safety and Justice
Contact for Students (Minor)	K.L.L. van Nunen
Intended for	The minor is multidisciplinary in nature, and is taught by lecturers from both the Faculty of Governance & Global Affairs at Campus The Hague (Leiden University), and from the Faculty of Technology, Policy & Management of TU Delft. The courses take place in The Hague and Delft, respectively. When admitting students, priority will be given to bachelor students from Leiden University, TU Delft and a limited number of students from Erasmus University Rotterdam. If there are still places available, students from other universities will also be admitted. The minor enables students to broaden their horizon, working together on security-related topics with colleagues from other universities.
Gives access to	N/A
Prerequisites Minor	To be admitted, students have to have obtained a minimum of 60 ECTS for their regular bachelor. Any foreknowledge of the issues discussed in the minor SSJ is not necessary, but students are expected to have some affinity with security issues. For the course on Security & Organisation, students are expected to understand the basics of statistical analysis.
Minor Exit Qualifications	The SSJ minor forms a good basis for the master in Crisis and Security Management at the faculty of Governance and Global Affairs, Leiden University, and the Safety and Security specialisation at the faculty of Technology, Policy and Management, TU Delft. However, successful completion of the minor does not give an automatic access to these master programs.
Minor Coherence / Goal	How do we protect our society against terrorism? Can we secure the complex world of cyberspace? Are self-driving cars safe enough? Who is responsible if something goes wrong? How do we balance surveillance and privacy? To provide this broad spectrum of knowledge to students, TU Delft and Leiden University set up a cooperation in the minor Security, Safety and Justice (SSJ). Security covers a broad range of research fields encompassing criminology, antiterrorism studies, and security management; the safety theme contributes via insights from risk management; and justice provides important concepts from legal studies, history and philosophy. Knowledge of essential concepts from all these fields is necessary to understand and analyse today's security and safety issues. This unique blend of knowledge from social and technical sciences allows students of the minor SSJ to address and analyse current security-related topics.
Minor Content 1	After completion of the minor, students will be able to: 1. Recognize safety, security and justice aspects in contemporary issues, and be able to synthesize scientific arguments for their presence. 2. Form an opinion on SSJ issues, which is grounded in scientific analysis and argumentation. 3. Apply basic techniques of analysis to contemporary SSJ issues. These techniques include: system analysis, risk analysis and perpetrator analysis. 4. Assess the value of scientific publications and use them in the analysis of SSJ issues, and be able to find publications that give deeper insight in these contemporary issues. 5. Recognize the role that SSJ issues have in their own study major, describe this role and explain it to others.
Education Methods	Study material. Most of the material for the SSJ minor is offered via Brightspace. For all six courses from the SSJ minor so both the courses offered by Leiden University as by TU Delft the Brightspace platform from Leiden University will be used. More information can be found in the individual course descriptions, found online. Course structure. The minor courses consist of a mixture of regular classes and work groups. Outside of the regular course hours, students will be expected to work on assignments and papers. More explanation can be found in the individual course descriptions.
Minor Assessment	Every course is evaluated separately. The minor is completed successfully if for all courses a grade of 6.0 or higher is obtained. Final grades are rounded off to the nearest half point at Leiden University and are rounded to the nearest tenth at TU Delft, except between 5 and 6. At both universities, for grades between 5.0 and 6.0, a 5.5 and higher will be rounded up to a 6.0, and grades below 5.5 will be rounded down to 5.0. So no final grades between 5.0 and 6.0 will be given. None of the individual constituent grades may normally be lower than 5.5, which means that as a rule no compensation is allowed, unless the rules for compensation are announced in the e-Prospectus / Study guide before the start of the course. If the final score is a passing grade, students are not allowed to retake to obtain a higher grade, unless given explicit permission by the Board of Examiners. More information on the exact forms of evaluation can be found online in the individual course descriptions. No matter what academic institution a student comes from, the SSJ minor falls under the course and examinations regulation of the bachelor Security Studies of Leiden University. These course rules are further expanded on in the rules and regulations of the Board of Examiners of the Institute of Security and Global Affairs. The Board of Examiners is also the relevant body for disputes. The SSJ minor consists of a coherent package of courses designed to help you build your knowledge in a logical and connected way. The six courses that together form the SSJ minor can only be taken within the framework of participation in the SSJ minor, so it is not possible to follow single courses. Furthermore, be aware that the SSJ minor needs to be completed in one academic year. It is not possible to retake courses that you have not past during the following academic year. If you think an exception to this rule applies to you, please contact the board of examiners of the Institute of Security and Global Affairs.
Maximum number of participants	140 (60 TU Delft, 60 Leiden University, 20 EUR)
Minimum number of participants	30
Minor Remarks / Schedule	The SSJ minor is offered jointly by Leiden University and TU Delft. This is reflected by the fact that half of the courses are given by TU Delft lecturers, in Delft, and the other half by Leiden lecturers, in The Hague. All locations are easily accessible by public transportation. For more information and registration details see https://www.tudelft.nl/en/tpm/education/minors/security-safety-justice-ld/ .

TBM025A	Security and Technology	5
Module Manager	K.L.L. van Nunen	
Instructor	K.L.L. van Nunen	
Co-responsible for assignments	Prof.dr.ir. G.L.L.M.E. Reniers	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Our current society is intrinsically linked with and dependent on technology. Safety and security threats are often associated with these technologies, such as earthquakes associated with gas extraction, the dangers of nuclear waste, drones, self-driving cars, or cybercrime on the internet. This causes society to be constantly exposed to risk. Often, the risk level only becomes known after the technologies have been deployed and undesired events occur. Contrary to relatively well-understood risk in organisations, the introduction of new technologies in society may imply uncertain and ambiguous safety and security risk, and may give rise to heavy political debate. In this course, you will learn to identify possible safety and security threats associated with new technologies, and provide suggestions for responsible design and deployment. In addition, you will learn to investigate side-effects of technologies aimed specifically at increasing security or safety, such as surveillance technologies and their effects on privacy.	
Study Goals	After this course, you are able to: 1) Explain the goals and methods of technology assessment, and apply it to a specific case 2) Identify and analyse possible security and/or safety implications of a (new) technology 3) Analyse how (new) technologies can be used to improve security and/or safety, and identify and analyse possible side-effects such as the effect on other values 4) Provide suggestions for responsible design and deployment of a new technology with respect to security, safety, and other values 5) Write a technology assessment report in group	
Education Method	Lectures, exercises and group discussions Writing a report in group	
Literature and Study Materials	Will be made available via Brightspace	
Assessment	An individual, on campus exam counts for 50% of the total grade. This exam will consist of several open-ended questions which need to be elaborated in a restricted time period. In case of COVID-19 restrictions, the exam will be a take-home exam. In normal circumstances, the exam will be an on-campus exam. A technology assessment report written in groups of 5 students counts for 50% of the total grade. Buddycheck will be used to follow-up the individual contributions to the group report. Students will have to review themselves and their teammates on several dimensions such as initiative, communication, attitude and complying with agreements. Actions will be taken if there are students with overall low scores (both of the self-assessment and the assessment by the peers), or if there are large differences between the self-assessment and the average given by the peers. If needed, an adjustment of the grade will be pursued. A resit in block 3 consists of rewriting the technology assessment report (at least 50% new text). In order to count in the final grade, the grade on the individual exam has to be 5 or more out of 10, and the grade on the group report has to be 5 or more out of 10. In order to pass the course, the mean of both assessments (individual exam and group report) has to be at least 5.8 out of 10.	
Enrolment / Application	This course is part of the minor Security, Safety and Justice, taught at TU Delft by a lecturer from TU Delft. The course can only be taken within the framework of participation in the minor SSJ. This minor is a consistent package of courses designed to help you build your knowledge in a logical and connected way. It is not possible to follow single courses. Furthermore, be aware that the SSJ minor needs to be completed in one academic year. It is not possible to retake courses that you have not past during the following academic year. If you think an exception to this rule applies to you, please contact the board of examiners of the Institute of Security and Global Affairs.	
Category	BSc level	

TBM027A	Security Integration Project	5
Module Manager	Dr. M.L.C. de Bruijne	
Instructor	Dr. M.L.C. de Bruijne	
Co-responsible for assignments	K.L.L. van Nunen	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	3	
Course Language	English	
Required for	Minor Security, Safety and Justice	
Expected prior knowledge	This course is part of the minor Security, Safety and Justice, taught by a lecturer from TU Delft at TU Delft. The course can only be taken within the framework of participation in the minor SSJ.	
Course Contents	The course Security Integration Project represents the culmination of the minor Security, Safety and Justice. The knowledge students have obtained during previous courses within the SSJ minor is varied and diverse. This course has as its particular goal to synthesize and integrate the knowledge that has been obtained throughout the minors and challenges students to apply the different theories, approaches, and methods from SSJ courses to a real case. Students are encouraged to analyse an earlier well-researched safety and/or security incident which took place in a multi-actor environment and analyse a specific multi-actor related aspect and apply knowledge they have gained in the SSJ (and if possible from their study discipline). Students are encouraged to reflect upon whether and how the knowledge obtained can produce practical insights.	
Study Goals	The Integration course at the end of the Minor is designed to help students to synthesize SSJ knowledge and skills. In addition, the course is designed to assess the students ability to integrate and apply theoretical and/or methodological knowledge on SSJ and to test their ability to apply SSJ knowledge and skills effectively, within the context of a real world case. After this course you: - have insight in the impact of a multi-actor setting on a safety and/or security-related problems. - can analyse and synthesize knowledge from safety and/or security incident reports for research. - can apply the knowledge and methods taught within the SSJ minor to a real case. - have basic knowledge with regard to the execution and design of case study research - can design and conduct a (small) qualitative case study into a security incident. - can work and communicate effectively with students from other disciplinary backgrounds on designing and conducting a (small) joint security related research project.	
Education Method	Lectures Individual and Small group exercises/assignments Group discussions Participation in a small crisis decision making simulation (when circumstances permit) Participation in lectures, discussions and exercises is required in order to obtain a grade. All assignments/exercises have to be fulfilled.	
Literature and Study Materials	Will be made available via Brightspace	
Assessment	An individual literature review (30%) with a minimal grade of 5. Paper written in multidisciplinary groups (70%) with a minimal grade of 5. Individual contributions should be listed in an appendix. Problems with group cooperation should be reported as they occur and will be taken into account if necessary. Buddycheck will be used to follow-up the individual contributions to the group report. Actions will be taken if there are students with overall low scores (both of the self-assessment and the assessment by the peers), or if there are large differences between the self-assessment and the average given by the peers. If needed, an adjustment of the grade will be pursued. In order to pass the course, the combined grade of both assessments (individual literature review and group report) has to be at least 5.8 out of 10. A resit consists of rewriting the group report (at least 50% new text) in block 3.	
Enrolment / Application	This course is part of the minor Security, Safety and Justice, taught at TU Delft by a lecturer from TU Delft. The course can only be taken within the framework of participation in the minor SSJ. This minor is a consistent package of courses designed to help you build your knowledge in a logical and connected way. It is not possible to follow single courses. Furthermore, be aware that the SSJ minor needs to be completed in one academic year. It is not possible to retake courses that you have not past during the following academic year. If you think an exception to this rule applies to you, please contact the board of examiners of the Institute of Security and Global Affairs.	
Category	BSc level	

TBM0822TU	Engineering risk management in organisaties	5
Module Manager	Prof.dr.ir. G.L.L.M.E. Reniers	
Instructor	K.L.L. van Nunen	
Instructor	Dr. M. Yang	
Co-responsible for assignments	K.L.L. van Nunen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course will discuss risk in a broad sense. Based on statements and myths, a discussion is initiated on a topic related to engineering risk management in organisations. The students are invited and urged to think together with the lecturer on risk issues. Risk is conceptualized as a topic of uncertainty and possible loss from different perspectives (safety, security). Topics are further elaborated and theories/models/metaphors are taught while further discussing.	
Study Goals	The purpose of this course is to be able to understand risk in all its facets, to discuss about the concept and also to apply the knowledge and the attitude in applications. The student should be able to process operational risk knowledge (models, attitudes, concepts, etc.) into understandable and practical info and sound reasoning. The student should be able to link risk with safety and security issues and be able to reflect and debate on a variety of topics.	
Education Method	The course will be taught via interactive discussion lectures. Before each lecture, there are mandatory preparations such as reading book chapters and/or articles. There can be some exercises in group when needed.	
Assessment	The course content will be tested by an individual, on campus exam (closed book). This exam will consist of an essay question which needs to be elaborated in a restricted time period. Furthermore, be aware that the SSJ minor needs to be completed in one academic year. It is not possible to retake courses that you have not past during the following academic year. If you think an exception to this rule applies to you, please contact the board of examiners of the Institute of Security and Global Affairs.	

Year	2022/2023
Organization	Technology, Policy and Management
Education	Minors WM

WM-Mi-101-22	
Responsible Program Employee	G.M. Wakeling
Minor Coordinator	G.M. Wakeling
Administration by the Faculty of	TBM
Minor Title	Minor International Entrepreneurship & Development
Contact for Students (Minor)	Gareth Wakeling (minor coordinator) G.M.Wakeling@tudelft.nl
Intended for	Undergraduate students interested in technology-based entrepreneurship within a sustainable development context. The minor is an opportunity to apply entrepreneurial competencies to a real-life challenge within mainly the Global South.
Gives access to	NA
Prerequisites Minor	Have full-time availability for the program (so NO re-sits planned for semester 1). Being selected for the minor through the application procedure. This minor is open for English speaking students. This minor should be your first choice. The minor has its own exam dates and deadlines and it is not possible to resit the minor.
Minor Exit Qualifications	This minor program allows students to work abroad to develop technical solutions for complex challenges that will contribute to socio-economic development. During the minor, you will work in multi-disciplinary teams to solve a challenge that has been assigned to you by a project provider. The majority of these challenges are located within developing and emerging markets and focus on pro-poor growth. Together with your team, you will learn how to manage and work on a technology-related project in a different cultural and institutional setting.
Minor Coherence / Goal	The overall learning goal of the minor is to set up and manage entrepreneurial projects that contribute to sustainable development in the Global South. The following sub-learning goals will contribute to this overall learning goal through the various minor courses. 1. To assess and develop individual and group level entrepreneurial skills, knowledge and attitudes through the application of entrepreneurship models and theories in preparation for the Intercultural Integration Project 2. To co-create a project brief for an entrepreneurship project commissioned by external partners of the minor International Entrepreneurship & Development 3. To develop a well-founded research project plan by systematically implementing research methods and analysis techniques to be prepared for upcoming field research 4. To apply diverse theoretical tools and entrepreneurial competencies as taught within the minor courses to solve complex socio-technical challenges within a local context. 5. Conduct a feasibility report on the preconditions of a sustainable and impactful project and how it contributes to development in the country's context
Minor Content 1	The minor program is structured as follows: In the first two months, students will complete a set of three courses to help prepare them for their time abroad. These courses will help students adopt an entrepreneurial mindset and prepare them to work effectively in different cultural contexts. Students will also learn how to conduct research and learn about the local ecosystem and stakeholders of their challenge. Finally, they will prepare a project plan for their time abroad. In the second phase of the minor, if health and safety restrictions permit, students will travel abroad. During their time abroad, students are required to adopt an entrepreneurial mindset as they research, develop and execute a solution to a challenge. Challenges providers include startups, non-governmental organisations, (social) enterprises, universities or governmental organisations. At the end of the minor, students will assess the impact and feasibility of their solutions and reflect on the competencies they have developed as a team and as individuals.
Education Methods	Lectures, workshops and field work
Minor Assessment	There are 4 minor courses in total. 1. Q1 Entrepreneurial Thinking (4 EC) 2. Q1 Preparations for Intercultural Research Project (7 EC) 3. Q1 Development, Sustainability and Impact (4 EC) 4. Q2 Intercultural Integration Project (15 EC) Every minor course will be assessed separately. It is impossible to re-sit this minor as it involves group work and long-distance travel.
Maximum number of participants	60 Selectie Selectie vindt plaats door middel van een motivatiebrief die ingaat op de volgende selectiecriteria: motivatie om aan de minor mee te doen kennis van de ontwikkelingsproblematiek actieve, ondernemende houding ervaring in omgang met culturele diversiteit ervaring met samenwerken in een team neven-/vrijwilligers activiteiten cijfergemiddelde en studieritme (nominaal of niet) De brief van niet meer dan 700 woorden dient voor 1 juni naar de minorcoördinator opgestuurd te worden.
Minimum number of participants	30
Minor Remarks / Schedule	Above all, this program is prone to the ever-changing global environment. Especially with circumstances such as the coronavirus we need to prioritise health and safety which might mean that you cannot travel to the country or region where your project is located. Instead, we will then adapt by offering a virtual project in the Global South or an alternative assignment in The Netherlands. Such changes are at the discretion of the teaching staff.

TBM013B	Entrepreneurial Thinking	4
Module Manager	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	The course is only available to students that take part in the minor International Entrepreneurship and Development, e.g. Minor WM-Mi-101-21.	
Course Contents	What does it take to be an entrepreneur? An entrepreneurial mindset is one that is able to manage risk, identify opportunities and finally, innovate, develop and execute ideas that are commercially viable and feasible. In the course Entrepreneurial Thinking we will explore different theories and practices that contribute to your ability to think and act more entrepreneurially. More specifically, we will look at individual entrepreneurial traits, team composition, effective management of entrepreneurial processes, conducting riskiest assumption tests, and the desirability, feasibility and viability of a proposed idea. It is your job to explore how the theories discussed and practiced in class can help you enhance your entrepreneurial capabilities at an individual and group level. The experiences and lessons learnt in class will ultimately help you to develop and execute more effective solutions for a respective opportunity. As such, it is expected that you eventually apply what you have learnt in this course to your internship period in the Global South.	
Study Goals	The overarching objective on this course is: to assess and develop individual and group level entrepreneurial skills, knowledge and attitudes through the application of entrepreneurship models and theories in preparation for the intercultural internship. To achieve this, you will be able to do the following at the end of this course: Learning objective 1: To apply agile and human-centred frameworks to the process of creating complex, sustainable, and technical products and services within a local development context Learning objective 2: To conduct riskiest assumptions tests that will rapidly test, measure and build evidence-based project or start-up solutions within a local development context Learning objective 3: To form effective startup teams that optimally utilise the entrepreneurial competencies of each team member Learning objective 4: To develop desirable, feasible and viable value propositions for a product or service in a local development context Learning objective 5: To translate project ideas into business models that are financially and commercially robust, environmentally conscious and socially impactful.	
Education Method	A combination of lectures, exercises, reading assignments and in-class discussions.	
Assessment	During this course you will be assessed at the individual and at the group level. The individual assessment consists of two components. A personal log book and an essay. This counts for 30% of your final grade. The group assessment consists of three components. A start-up plan based on the theories discussed in class, conducting real assumptions tests and documenting the results, and finally a group presentation where you will explain how the learnings of this course will be applied to your internship period. This counts for 70% of your final grade.	

TBM014C	Preparations for Intercultural Research Project	7
Module Manager	N.Y. Aydin	
Instructor	E. van Andel	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course aims to prepare its participants in designing their intercultural research projects as first-time researchers as part of the minor International Entrepreneurship and Development. The course focuses on various research approaches, methods, and techniques for the participants to conduct research projects within the minor International Entrepreneurship and Development. The course provides a set of lectures, including research methods, project management, conversational interviews, reading assignments, and in-class discussions combined with group assignments. The participants will work with their project team and reflect on various research approaches, methods, and techniques in workshop sessions which are the main activities throughout the course. The participants will work with their project team and participate in coaching sessions and do assignments that help them prepare for the upcoming internship project in Q2. At the end of this course, the participants will produce a report and a presentation based on the result of the intercultural research project for their upcoming field trip.	
Study Goals	After successfully completing the course, the participants will be able to: 1) Understand the principles of the selected research approaches and methods; 2) Reflect on the strengths and limitations of the research methods and analysis techniques; 3) Create a research project plan with substantial research methods, finance strategy and risk and safety plan for the upcoming intercultural field research. 4) Set up at least 5 expert consultation meetings to prepare and reflect on the upcoming field project; 5) Design and present a detailed project research plan for the upcoming intercultural internship.	
Education Method	The course will follow a blended education technique including a mix of lectures, discussions, and pre-class reading assignments about research methods, and EDx lectures.	
Assessment	Students (in groups) will be evaluated based on: 1. Expert meetings and supervisor meetings (10%) 2. Project Research & Safety Plan (55%) 3. Presentation of the project plan (30%) 4. Participation (5%)	

TBM029A	Development, Sustainability and Impact	4
Module Manager	Dr.ing. V.E. Scholten	
Instructor	F. Delgado Medina	
Responsible for assignments	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. T.L. Dolkens	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	The course is only available to students who participate in the minor International Entrepreneurship and Development, e.g., Minor WM-Mi-101.	
Course Contents	The core objective of this course is to prepare the students to be able to build up a sustainability-oriented mindset for their internship project and country's contexts in terms of Development (Macro-level of analysis), Sustainability (Meso -level of understanding), and Impact insights (Micro-level of implementation) to contribute to the feasibility of their prospective project. The feasibility of the project responds to the meanings of a successful project in a developing country and its feasible contribution to sustainable and social contribution regarding the practical implementation of the concept and implementation tools and approaches that will help students to address their project challenges concerning "Development, Sustainability, and Impact." The students will work in groups, each of which shall participate in, and assess, workshop presentations and a final feasibility report based on the analysis and implementation of the tools covered in this course within their upcoming field trip. The final report also contributes to the overall LO of the course, which is to conduct a feasibility study on the preconditions of a sustainable and impactful project and how it contributes to development in the country's context.	
Study Goals	By the end of this course, you should be able to: LO1: Conduct a feasibility report on the preconditions of a sustainable and impactful project and how it contributes to development in the country's context. LO2: Discuss and define how your project contributes to the development and how innovation strategies can tackle global challenges in the country's setting project. LO3: Analyze how your project will encounter or not sustainability principles in the local context. LO4: Assess and improve how your project fits into the entrepreneurship ecosystem. LO5: Improve stakeholder management in your project context. LO6: Enhance your project's contribution to creating social and sustainable impact.	
Education Method	A merger of lectures, guest lectures, work-group activities, reading assignments, and in-class discussions	
Assessment	The grading for this course will be based on: group presentation and critical summary (30%), feasibility report (70%).	
	More information can be check in the course overview at Brightspace.	

TBM030A	Intercultural integration project	15
Module Manager	Dr.ing. V.E. Scholten	
Instructor	E. van Anandel	
Instructor	F. Delgado Medina	
Instructor	G.M. Wakeling	
Co-responsible for assignments	Dr. T.L. Dolkens	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Course Contents	The 15 ECTS Q2 module has been designed to allow students to practise and reflect on the theories, skills and tools learned during the first half of the minor. Assuming no safety restrictions, students will travel to complete a practical 2-3-month project hosted by an NGO, startup, (social) enterprise, university or governmental organisation, usually in the Global South. The academic focus during the module is applying entrepreneurial and project management skills within a challenge-based and experiential learning pedagogy. In doing so, students will work closely with project stakeholders to understand and consider the local context to develop technical solutions that contribute to sustainable socio-economic development. Each project has a (local) challenge owner and is co-supervised by a member of staff from TU Delft. Students will work in multi-disciplinary teams to solve the challenge they have been assigned.	
Study Goals	After completing the intercultural integration project, students will be able: - To develop and execute a solution to a complex socio-technical challenge as presented by their project provider. - To incorporate entrepreneurial competencies and project planning techniques - To manage communication with local stakeholders and the transfer of technology in a culturally sensitive manner. - To improve and reflect on one's own entrepreneurial competencies and role within a startup team - To assess the effectiveness of the project outcomes and present this to a panel of judges made up of experts or peers, and a final project group report	
Education Method	Before the start of the internship, the student groups should start the process of fine-tuning and understanding the challenge they will solve during their time abroad. This process will be facilitated through the other minor courses before a student leaves for their internship. Together with their project provider and TU Delft supervisor, students will give further shape to the challenge and develop a project plan they will aim to execute during the internship. During the project work, students will have regular coaching sessions with their TU Delft supervisor and work actively to identify and present a sustainable solution to their challenge. The final group report should assess the impact and feasibility of their solutions and reflect on the competencies they have developed as a team and as individuals.	
Assessment	Assessment and grading of the final group report are carried out by the TU Delft academic supervisor in consultation with local project providers. A 360-degree peer assessment of individual contributions to the group project and teamwork dynamics may be reflected in the final grade individuals receive.	
Remarks	The module takes place in the form of a three-month stay abroad to complete a challenge or project for an external organisation or company. Projects are pre-selected by the minor coordinator and the minor teaching team to deliver the same learning outcomes in line with the goals of the IED minor. Once accepted into the minor, students will be matched with a project based on academic background, skills required for the project, and, as far as possible, personal preferences and motivation will be taken into account.	

Year	2022/2023
Organization	Technology, Policy and Management
Education	Minors WM

WM-Mi-180-22	
Program Coordinator	Mandi Astola - minor coordinator
Contact for Students	stuco-ba-kg@hum.leidenuniv.nl

TBM001B	Introduction to Responsible Innovation	3
Module Manager	Dr. M. Astola	
Instructor	Dr. M. Astola	
Responsible for assignments	Dr. M. Astola	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	2/0/0/0 (3,5 hours home work)	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This is the introductory course for the Responsible Innovation minor. Technology plays an ever-increasing determining factor in our lives as individuals and society at large. New technological innovations contribute to human well-being, but they may also introduce considerable risks to humans, animals, the environment and future generations. As such, we have every reason to insist that the new technologies we develop as a society respect the values we hold dear. Various governments, companies and research funding agencies have now recognized this need for responsible innovation. The main aim of this course is to reflect on the nature of technological innovation in relation to societal and ethical concerns, and to gain a solid understanding of the notion of responsible innovation and value-sensitive design. Responsible innovation can, as a concept, be understood in a substantive and in a procedural sense. As a procedural notion, responsible innovation refers to a process of innovation that meets certain norms, like transparency, public engagement, and accountability (to stakeholders and to society). As a substantive notion, responsible innovation refers to a process of innovation that results in innovative technologies that reflect important moral values, including health, safety, human and animal welfare, sustainability, justice, inclusiveness, democracy, privacy, trust, and autonomy. In this introductory course we will learn about responsible innovation by studying these concepts and by focusing on some of the above-listed values, asking just how they can and should be implemented in our innovations. Throughout the course, we will explore questions such as: what do we mean by good or responsible design and innovation? What is the role of technology in our daily lives, and what is the relationship between moral values and innovation? And how (if at all) can we really anticipate the future socio-ethical implications of newly emerging technologies?	
Study Goals	After following this course, students should be able to: Understand what responsible innovation means by explaining the following key theoretical concepts: ethical anticipation, ethical values, the value-laden nature of technological artefacts, procedural responsible innovation, substantive responsible innovation, value sensitive design. Analyze and debate ethical values at stake in concrete cases of innovation. Identify at least one methodology for embedding ethical values into technological artefacts. Discuss challenges with respect to the embedding of ethical values in technological artefacts Develop a cogent ethical argument applying course theory to a case study of technological innovation.	
Education Method	The education method of this course is mixed: meetings will include video lectures by the instructor, interactive group discussions and debates. There may be guest lectures as well, depending on availability of the relevant speakers.	
Assessment	Group Assignment (value sensitive design 'tinkering workshop' + presentation) (40%) Individual Assignment (40%) (weekly written reflections) Take home examination(20%) to promote an inclusive learning environment, students are able to propose an alternative format for the individual assignment that best matches their learning style: e.g. instead of weekly written assignments they could opt for a podcast or wiki-page format. The requirements in terms of covering theoretical content will be the same for each of these assignments and the student and instructor must come to an agreement in advance before the student can pursue this alternative format. In the event of unforeseen circumstances, e.g. COVID-19 related measures, the in-class assessment moment (tinkering workshop) will be replaced with an additional take-home assignment	

TBM002A	Student Project Group 1	6
Module Manager	Dr. M. Astola	
Instructor	Dr. M. Astola	
Responsible for assignments	Dr. M. Astola	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	This course is only open to students enrolled in the minor Responsible Innovation.	
Course Contents	The integrative focal point of the minor Responsible Innovation is the Student Project Groups (SPGs) which are mixed groups of 4-6 students coming from the three universities, tutored by academic coaches. Most projects are done in cooperation with a company or societal organisation, and during the process each SPG is in touch with this target stakeholder. In the SPG, the student will apply the Intervention Cycle to a real-life case of innovation. At the end of each quarter, there will be a plenary seminar where the groups will present their results to fellow students and coaches and, if feasible, also the target stakeholders. The SPGs are formed during the first integration seminar, also known as the Kick-off day of the minor. In addition, students will attend several skill labs. Attendance of the skill labs are mandatory.	
Study Goals	At the end of the learning experience you can demonstrate understanding of the intervention cycle by applying the framework to a specific case of responsible innovation by working in an interdisciplinary team. This includes: 1. Applying practical and theoretical knowledge to papers and presentations; and 2. Reflecting on teamwork and team decision making processes.	
Education Method	The structure of the SPGs is based on the six steps of the Intervention Cycle: Problem Identification, Problem Analysis, Problem Definition, Solution (Re)Design, Solution Implementation and Solution Evaluation. Students will cover the entire cycle during each of the two quarters. The results will be presented during the plenary sessions at the end of the first and second quarter.	
Assessment	Report and presentation	
Tags	Ethics	

TBM003A	Student Project Group 2	6
Module Manager	Dr. M. Astola	
Instructor	Dr. M. Astola	
Responsible for assignments	Dr. M. Astola	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is only open to students enrolled in the minor Responsible Innovation.	
Course Contents	The integrative focal point of the minor Responsible Innovation is the Student Project Groups (SPGs) which are mixed groups of 4-6 students coming from the three universities, tutored by academic coaches. Most projects are done in cooperation with a company or societal organisation, and during the process each SPG is in touch with this target stakeholder. In the SPG, the student will apply the Intervention Cycle to a real-life case of innovation. At the end of each quarter, there will be a plenary seminar where the groups will present their results to fellow students and coaches and, if feasible, also the target stakeholders. The SPGs are formed during the first integration seminar, also known as the Kick-off day of the minor. In addition, students will attend several skill labs. Attendance of the skill labs are mandatory.	
Study Goals	At the end of the learning experience you can demonstrate understanding of the intervention cycle by applying the framework to a specific case of responsible innovation by working in an interdisciplinary team. This includes: 1. Applying practical and theoretical knowledge to papers and presentations; and 2. Reflecting on teamwork and team decision making processes.	
Education Method	The structure of the SPGs is based on the six steps of the Intervention Cycle: Problem Identification, Problem Analysis, Problem Definition, Solution (Re)Design, Solution Implementation and Solution Evaluation. Students will cover the entire cycle during each of the two quarters. The results will be presented during the plenary sessions at the end of the first and second quarter.	
Assessment	Report and presentation.	
Tags	Ethics	

TBM015A	Responsible Management of Risk, Safety and Security	3
Module Manager	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Prof.dr.ir. B. Taebi	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Minor Responsible Innovation	
Course Contents	Introduction to the field of risk and safety/ security based on various texts (book, scientific papers and book chapters). In the course, safety and security are considered from multiple perspectives (industrial safety, transport safety, organisational safety, societal safety, occupational safety, fire safety, structural safety, physical security, cyber security). Methods and techniques will be taught to model the factors which influence safety and security in these domains (technical-, human-, and organisational factors), by adopting a decomposition approach of the overall socio-technical system. Methods and techniques of safety-by-design and security-by-design principles for systems and processes are taught to inherently reduce risks of failure during the life cycle of these systems and processes. Examples of fault-tolerant design, fail-safe design, fool/huffer-proof design in the above domains are discussed.	
Study Goals	In this course, students will be introduced to safety and security risk management as it is applied in multiple domains. The focus is on how responsible decisions on safety and security, using the concepts of risk, can be made, in particular in the design phase of systems and processes. Students learn to understand and use the concept of risk in safety and security settings, being aware of its benefits and limitations. They acquire practical knowledge and skills that allow them to describe the safety and security in qualitative and quantitative terms, to investigate the factors which have an impact on safety and security, and to derive and analyse decisions on safety- and security measures, made by risk managers. The course starts with safety risk first (accidents), and then extends its scope to security risks (threats and attacks). The following topics will be covered: Introduction of important safety and risk concepts, such as probability, frequency, hazard/ threat, vulnerability, impact/ consequence Short history of safety and risk thinking The management of risk and safety in public and private organisations Similarities and differences between safety and security risks Quantitative and qualitative assessment of safety and security risks Several risk analysis techniques such as fault/ attack trees, hazardbarriertarget HBT-model and bowties The psychology of safety and security, human error A systems view on safety/ security The principles of safe-by-design and security-by-design During the course various reports on risk/ safety topics in media will be discussed. After this course, students are able to: Understand, apply, and quantify risk and related concepts in safety and security settings, as well as reasoning about these Explain the similarities and differences between safety and security risks and their assessment Understand the tasks and responsibilities of the safety/ risk manager or professional Apply several safety and security risk analysis techniques such as fault and attack trees Make well-founded decisions regarding safety or security issues and countermeasures, explain these choices and communicate them Apply the safe-by-design and security-by-design principles	
Education Method	Plenary lectures (mandatory) Exercises (carried out in small groups of 4-5 students)	
Assessment	#### Assessment One group assignment, to be published in a written report, and to be presented in a 30 min. presentation, followed by Q&A with the teaching staff and students. Closed-book exam (individual). For students following the Minor 'Responsible Innovation', student attendance and participation is required and all assignments/exercises have to be fulfilled in order to receive a passing grade. Only one lecture can be missed. If a second lecture is missed, an assignment will be given to compensate. If a third lecture is missed, a passing grade can no longer be obtained, unless permission is granted from the Board of Examiners (with consent of the Course Coordinator). #### Weighing 40% for the group assignment and 60% for the individual exam. #### Resit The students are allowed one re-sit per examination. It is not allowed to re-sit an examination or assignment for which they have received a pass (6,0 or higher). It is allowed to re-sit an examination or assignment which they haven't done during the first occasion. The re-sit format needs to be discussed with the teacher of the course in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit has to take place within study year 2021-2022. This means the students have to complete the minor within one study year.	

Year	2022/2023
Organization	Technology, Policy and Management
Education	Minors WM

WM-Mi-191-22	
Minor Coordinator	M.B.O.T. Klenk
EC Program	30 ECTS.
Prerequisites	2 jr. BA.
Introduction 1	The Minor Companies and Innovation focuses on new (high-technology) product innovations within established companies, i.e. internal venturing or corporate entrepreneurship, by examining their potential values from diverse aspects such as economics, management, strategy, ethics, safety and risk. The requisite view is that companies are not isolated entities but are embedded in a social context with various values and expectations. Firstly, there are the markets a company is operating in, where the laws of economics rule: supply and demand, investment, financing and marketing. Secondly, there are restrictions (and opportunities!) set by the enforced rules of the law. For instance, the legislator may demand competition and forbid exclusive price arrangements requiring firms to think of their strategies, such as their intellectual property protection. Thirdly, there are considerations regarding safety or security and risk, that is, the safety (or security) of the product(s) the company produces, but also the safety of the employees involved in the production process and environmental safety. Finally, current opinions about what is good and immoral (or what is responsible and irresponsible) determine the company's ethical standpoint shaping its corporate social responsibility. The ability to integrate technological development with diverse values of firm strategies, market, ethics and safety is key to innovations in companies. In sum, an integrated insight into these distinct viewpoints is essential for companies to make sensible decisions and to develop new products that are both technologically and socially relevant and acceptable. The minor will be concluded with a final presentation by students (based on their business plan) to a panel of experts from academia and industry. All courses are offered in English.
Program Goals	The aim of the Minor is to offer students insights into how economics, strategic management, ethics, and safety and risk considerations impact innovation. This emphasis on multidimensional assessment of technological innovations within companies make the minor relevant, attractive and challenging for enterprising students, particularly technology-oriented students who would like to enhance their knowledge of management of innovation.
Program Structure 2	Students work in (heterogeneous) groups of four or five on their business plan.
Fail or Pass Regulation	All courses will require a grade of 6 or higher for passing.
Administration by the Faculty of	TBM
Minor Title	Companies and Innovation
EC Minor (Volume)	30.
Minor Semester	First semester.
Expected prior Knowledge	None.
Education Methods	Each course will apply its own teaching methods. Detailed info can be found further in the flyer or minor's website.
Minor Assessment	All courses will be assessed separately, with either an exam or a group assignment. For the integration course, students work in groups on their business plan, which they will present at the end of the minor to a panel of internal and external experts.
Maximum number of participants	60 students.

TBM004A	Strategic Management Game	4
Module Manager	Dr. G. van de Kaa	
Instructor	Dr. G. van de Kaa	
Instructor	Ir. V.C.M. Sobota	
Instructor	F. Grillo	
Responsible for assignments	Dr. G. van de Kaa	
Co-responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course C&I Management Game focuses on formulating and implementing a strategy for large firms. Throughout the course, students will be acquainted with a variety of strategy tools which are utilized by strategy researchers and students will apply those perspectives to a real life situation (business strategy game).	
Study Goals	After the course, students are able to understand, analyze and conclude on companies strategy After the course students are able to formulate and implement a firms strategy	
Education Method	Teaching game (online course)	
Literature and Study Materials	teaching game available through www.bsg-online.com	
Assessment	Students will play an online game. The game consists of several quizzes and exams that form the total grade.	

TBM005B	C&I Integration Course	6
Module Manager	K.L.L. van Nunen	
Instructor	K.L.L. van Nunen	
Co-responsible for assignments	Prof.dr.ir. G.L.L.M.E. Reniers	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The objective of this integration course is to apply the knowledge and content of the minor 'Companies and Innovation' to a concrete case, i.e. an innovative (tangible) product for an existing company. In a group of five students, you will develop, describe and elaborate a technological development in a business model canvas (Q1) and a business plan (Q2), integrating the four perspectives (finance, management, safety, ethics) presented in the minor.	
Study Goals	In the first quarter, you will come up with an innovative (tangible) product in a group of five students. During part I of the course, you will elaborate this product using the business model canvas. In this business model canvas, you will list all the key building blocks to develop, manage, maintain, market and sell your product. The content of the other courses of the minor given during Q1 will serve as a foundation for the elaboration of your product. The deliverable at the end of the first quarter is the business model canvas, which makes up 15% of the total grade. The grade on the business model canvas is partially a group grade (80%) and partially an individual grade (20%). In the second quarter, the innovative product and its company will be further elaborated (in the same groups of students as in Q1). Financial, managerial, ethical, and safety related aspects will be included in a business plan. The content of the other courses of the minor (given during Q1 and Q2) will serve as a foundation for the business plan. The deliverable at the end of the second quarter is twofold: the business plan, which makes up 60% of the total grade, and a presentation and defense of the business plan for an expert panel, which makes up 25% of the grade. The grade on the business plan and the presentation is partially a group grade (80%) and partially an individual grade (20%). After taking this course you will be able to: - Apply the knowledge and content of the minor 'Companies and Innovation' to a concrete case, i.e. an innovative (tangible) product for an existing company - Recognise and identify new challenges and opportunities for companies to develop/innovate a new product - Compose a business model canvas and elaborate on its key building blocks to develop, manage, maintain, market and sell an innovative (tangible) product - Integrate the four perspectives presented in the minor (finance, management, safety, ethics), and analyse tensions between the different perspectives deriving from the innovation process	
Education Method	There will be weekly contact moments with the lecturer by means of the flipped classroom. This means that before the contact moment, you will prepare yourself, for instance by reading literature or making an assignment. During the contact moment, you will practice applying key concepts with feedback of the lecturer and your peers. After the contact moment, you will check your understanding and extend your learning. By means of weekly assignments, your progress will be evaluated. These contact moments will be supplemented by guest lecturers.	
Literature and Study Materials	Attendance and participation at all contact moments (flipped classroom) and guest lectures is mandatory. The individual grading will be based on the attendance and participation during the lectures. If more than two contact moments/guest lectures are missed (for Q1 and Q2 combined), a passing grade can no longer be obtained, unless permission is granted from the Board of Examiners (with consent of the module manager of the course).	
Assessment	- Supporting literature (scientific papers, book chapters, videos) will be made available during the lectures and via Brightspace - Slides from the (guest)lectures Quarter 1 The deliverable of C&I Integration course part I is the elaboration of the business model canvas. The business model canvas will count for 15% of the total grade of which: - 80% is a group grade (based on the quality of the deliverable) - 20% is an individual grade (based on the attendance and participation during the lectures) Quarter 2 The deliverable of C&I Integration course part II is twofold: - The elaboration of the business plan. This will count for 60% of the total grade. - Group presentation and defense of the business plan for an expert panel consisting of members of the business and academic community. This group presentation will count for 25% of the total grade. For both the business plan and the presentation, the grade is divided as follow: - 80% is a group grade (based on the quality of the deliverable) - 20% is an individual grade (based on the attendance and participation)	
Remarks	There will be weekly group assignments, which will not be quoted. However, these weekly group assignments have to be handed in on time, and have to meet certain quality standards in order to receive a passing grade for the course. This course is part of the Minor 'Companies and Innovation' and cannot be followed separately.	

TBM022A		Technology, Innovation and Ethics for Companies	5
Module Manager	M.B.O.T. Klenk		
Instructor	M.B.O.T. Klenk		
Responsible for assignments	M.B.O.T. Klenk		
Co-responsible for assignments	Dr. J.B. van Grunsven		
Contact Hours / Week x/x/x/x	0/2/0/0		
Education Period	2		
Start Education	2		
Exam Period	2 3		
Course Language	English		
Course Contents	Technology plays a critical role in our daily lives, in our organizations and institutions, and in society at large. New technological innovations contribute to human well-being, but they may also introduce considerable risks to humans, animals, the environment, and future generations. As such, we have every reason to insist that the new technologies we develop as a society respect the values we hold dear. Various governments, companies, and research funding agencies have now recognized this need for responsible innovation. The main aim of this course is to reflect on the nature of technological innovation in relation to societal and ethical concerns, and to gain a solid understanding of the notions of responsible innovation and value-sensitive design. Responsible innovation can, as a concept, be understood in a substantive and in a procedural sense. As a procedural notion, responsible innovation refers to a process of innovation that meets certain norms, like transparency, public engagement, and accountability (to stakeholders and to society). As a substantive notion, responsible innovation refers to a process of innovation that results in innovative technologies that reflect important moral values, including health, safety, human and animal welfare, sustainability, justice, inclusiveness, democracy, privacy, trust, and autonomy. In this introductory course we will learn about responsible innovation by studying these concepts and focusing on some of the above- listed values, as well as examining the role and responsibilities of companies in the development, usage, and proliferation of technologies and innovation. After all, those who seem most directly capable of making an impact on our technologically mediated world are, for better or for worse, not individual persons but collective agents in the form of companies. This leads us to explore questions such as: what do we mean by good or responsible design and innovation? How should companies deal with ethical issues and concerns? How should companies deal with the (unknown) risks of new technologies? What, if any, are the social responsibilities of private companies?		
Study Goals	At the end of the course, students should be able to: LO1. Identify various theoretical views regarding the nature of technology and its relation to ethical values. In particular, you should be able to identify and explain basic concepts from normative ethics and their relation to technological innovation. LO2. Explain various conceptions of responsibility and how they apply to companies. LO3. Think critically about whether and how values are embedded in technological innovations. LO4. Explain the meaning of the concepts of responsible innovation and value sensitive design. LO5. Implement the above-mentioned skills through interactive discussions and critical reflections on course readings. LO6. Analyse, develop, and write a cogent argument regarding the ethical aspects of innovation and moral responsibilities of companies.		
Education Method	A combination of online lectures, online discussions, take-home assignments (both individual and group) and online sessions. We will use MS Teams to communicate about the course, share educational materials, and notify students of deadlines. Please be prepared to participate using MS Teams.		
Literature and Study Materials	A reader will be made available on MS Teams shortly before the course begins.		
Assessment	There are three assessments for this course, all of which will be administered online: 1) a quizz, 2) reading responses, 3) and a group essay. The quizz accounts for 20% of the course grade, the reading responses as well as the group essay will each account for 40% of the final course grade.		

TBM023B	Innovation Management	5
Responsible Instructor	Dr. N. Pachos-Fokialis	
Module Manager	Dr. N. Pachos-Fokialis	
Instructor	Dr. N. Pachos-Fokialis	
Co-responsible for assignments	Dr. G. van de Kaa	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course discusses the challenges that companies face in introducing innovations and managing technological development, under the lens of the digital transformation of their key organizational functions. Firms may use innovation to improve their processes or to differentiate their products and services and outperform their competitors. However, the management of innovation is inherently difficult and risky: most new technologies fail to be translated into products and services, and most new products and services are not commercially successes. In short, innovation can enhance competitiveness, but it requires a specific knowledge and management style. The course adopts an integrative perspective in analysing the innovation management process and discusses how the organizations can search for innovative opportunities, select innovative projects, develop new products/services and capture a social, economic or business value from this process. It also discusses different types of innovation and defines critical elements for building an innovative organization and developing an innovation strategy. In sum, the course presents the main concepts and methods in innovation management and supports them with series of examples and applications from practice.	
Study Goals	By the end of the course, students should be able to: - Define innovation management as an organizational process, consisting of specific stages. - Identify different types and aspects of innovation. - Explain how the organizations can build innovative capabilities. - Discuss how the organizations search for innovation opportunities and explore innovative ideas. - Explain how the organizations select an innovation project, among a range of innovative ideas. - Analyse how the organizations develop a new product or service, and what strategies they implement in order to meet the customers needs and expectations. - Evaluate the innovation management process, taking into consideration the business, social and economic values of innovation.	
Education Method	The course is organized into 6 lectures and 2 presentation sessions. The first lecture will take place on 05/09. During the presentation sessions, student groups will present their group assignments. (More details will be announced at the first lecture)	
Literature and Study Materials	Textbook: Tidd, J., Bessant, J. R., & Pavitt, K. (2020). Managing innovation: Integrating technological, market and organizational change. Hoboken: Wiley (7th edition). Journal articles: Will be posted on Brightspace. Lecture slides: Will be posted on Brightspace.	
Assessment	The assessment of the course will consist of two parts: - Group assignment (50%) - Written exams (50%) For the purpose of the group assignment, students will work in groups of 5. Each group will select an organization and analyse how it manages innovation, in terms of search, selection, implementation and evaluation strategies. The written exams will include open-ended and multiple-choice questions and will take place on campus. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: - Group assignment (50%) - Timed take-home exam (open book) (50%) The partial grades for the group assignment and the written exams will be valid for the next academic year as well.	

TBM031A	Financing Innovation	5
Module Manager	M.B.O.T. Klenk	
Instructor	A.A. Ralcheva	
Co-responsible for assignments	Dr. G. van de Kaa	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Identifying and investing in innovation opportunities is key. In fact, organizations that do not invest in innovation struggle to keep pace in the rapidly changing business environment. However, even the best and most innovative ideas will go nowhere unless we can convince others to commit the necessary resources. That is why, finance can play a very important role in helping organizations and investors recognize the value in innovation.	
Study Goals	The Financing Innovation course will equip you with the necessary toolkit for making use of financial statements, performing financial statement analysis, estimating business value and making financing decisions from the perspective of an innovative company. Understanding the key techniques of financial valuation and capital budgeting used by organizations is critical for project owners (project managers or entrepreneurs) to make sure their ideas get the attention and funding they deserve. More specifically, you will gain experience building a financial model to assess a new product launch. We will consider key financial metrics that are often applied and learn which ones are reliable (and which are not). We will also look at how we can use the financial model to guide our attention as project managers in order to maximize the impact of our efforts. Along the way, we will gain insight into how financial markets work and how investors evaluate stocks and bonds. Finally, we will learn how to use these skills to understand the drivers of a company's value.	
	The Financing Innovation course can be seen as supportive to other fields such as strategy, marketing and innovation management. Often in these fields business decisions are formulated (e.g., in what markets or products to invest). In Financing Innovation we evaluate these choices and analyze how to fund them best.	
	The course is relevant for students planning to become entrepreneurs, student entrepreneurs, or students working in the high tech sector, consulting or public institutions supporting innovation and entrepreneurship.	
Study Goals	By the end of this course, you should be able to:	
Education Method	LO1: Discuss at least four types of financing available at different stages of the innovation process (e.g. accelerators, crowdfunding, business angels and venture capital).	
	LO2: Make use of financial statements for the purposes of evaluating firm performance and estimating firm value.	
	LO3: Apply a simple financial forecasting technique (such as the percentage of sales method) using Excel;	
Education Method	LO4: Estimate the value of financial assets such as stocks and bonds.	
	LO5: Apply at least three different capital budgeting techniques (e.g. NPV, IRR, Payback Period).	
	LO6: Explain the investment and financing decisions of (innovative) companies and their impact on the value of the firm.	
Education Method	This is a blended course which includes a combination of regular sessions, Excel workshops, cases, individual reading assignments, exercises and simulations to prepare you to build a valuation model for a hypothetical innovation project your final group assignment.	
Assessment	The regular sessions and workshops are on campus. However, depending on the COVID-19 measures, the regular sessions and workshops may be given through a hybrid form or through online and remote learning.	
	The course assessment is based on an individual written digital exam on campus and a group assignment. The exam is closed book. Only a calculator and a pen are allowed. Dictionary is not allowed.	
	The final grade for this course is determined as follows:	
Assessment	1. Individual Mobius exam on campus (60%)	
	2. Group assignment (40%)	
	In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows:	
Assessment	1. Remote individual Mobius exam (60%)	
	2. Group assignment (40%)	

TBM0821TU	Beheer van risico's en veiligheid	5
Module Manager	Prof.dr.ir. G.L.L.M.E. Reniers	
Instructor	K.L.L. van Nunen	
Instructor	Dr. M. Yang	
Co-responsible for assignments	K.L.L. van Nunen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course will discuss risk in a broad sense. Based on statements and myths, a discussion is initiated on a topic related to engineering risk management in organisations. The students are invited and urged to think together with the lecturer on risk issues. Risk is conceptualized as a topic of uncertainty and possible loss from different perspectives (safety, security). Topics are further elaborated and theories/models/metaphors are taught while further discussing.	
Study Goals	The purpose of this course is to be able to understand risk in all its facets, to discuss about the concept and also to apply the knowledge and the attitude in applications. The student should be able to process operational risk knowledge (models, attitudes, concepts, etc.) into understandable and practical info and sound reasoning. The student should be able to link risk with innovation and developing innovative products and processes.	
Education Method	The course will be taught via interactive discussion lectures. Before each lecture, there are mandatory preparations such as reading book chapters and/or articles. There can be some exercises in group when needed.	
Assessment	The course content will be tested by an individual, on campus exam (closed book). This exam will consist of an essay question which needs to be elaborated in a restricted time period.	

Year	2022/2023
Organization	Technology, Policy and Management
Education	Minors WM

WM-Mi-196-22	
Contact for students	minorfrugalinnovation@asc.leidenuniv.nl

TBM008B	Technology and Frugal Innovation	3
Module Manager	Prof.dr. C.P. van Beers	
Instructor	Dr.ir. J.C. Diehl	
Instructor	Prof.dr. C.P. van Beers	
Instructor	Dr.ir. R.E.F. Lindeboom	
Responsible for assignments	Prof.dr. C.P. van Beers	
Co-responsible for assignments	Dr.ir. R.E.F. Lindeboom	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	all students that are enrolled in the LDE Minor Frugal Innovation for Sustainable Global Development.	
Course Contents	This course is the first of the three thematic courses in the LDE Minor Frugal Innovation for Sustainable Global Development. Because students in this minor may not have a purely technological background, this course on Technology and Frugal Innovation will start from the point of view of innovation management by which students become more acquainted with innovation theories and how frugal innovation relates to and fits into these theories. From there the step is made to innovation for low resource or resource constrained settings, what do such settings imply for innovation and product development cycles, and the technologies involved. Students develop answers to questions such as: What is frugal technology and how does it come about? How can we use technology to address and overcome tensions between value capturing and value creation? Next, two strategic themes will go more in-depth and to which general concepts and notions are applied: 1) clean water and sanitation, and 2) medical diagnostics targeting good health and well-being (SDG 3). The first strategic theme is access to clean drinking water and sanitation. Access to clean drinking water is one side of the coin. The other side refers to water pollution due to the increasing degree of urbanization, which puts the natural hydrological cycle under great pressure. Discharge of municipal and industrial waste water increasingly threatens clean water reservoirs, which due to the growing water demand are already experiencing seasonal shortages in many areas in Southeast Asia and Sub-Saharan Africa. By means of practical cases students get acquainted with basic (frugal) concepts in water purification and hydrometry and/or weather forecasting. The connection of sanitation with the second strategic theme is also discussed. The second strategic theme focuses on good health and well-being. Researchers at TU Delft work together with designers, doctors and local parties on integrated frugal solutions. The aim is to design radically new diagnostic equipment that optimally functions within the specific characteristics of local health systems. This theme will take on the challenge to adapt the developments of new medical equipment to the needs and requirements of local healthcare systems, through case driven education.	
Study Goals	After this course students are able to: - Reproduce and interpret knowledge about frugal innovation from a technology perspective; - Apply this perspective critically to case studies from different parts of the world; - Analyse and interpret the potential of frugal innovation in the pursuit of the UN Sustainable Development Goals from a technology perspective; - Learn from other disciplines, in particular innovation theories and management as well as specific technology disciplines, and relate this knowledge to and integrate into their own disciplinary background, and - Make a constructive contribution to the debate on frugal innovation and sustainable global development.	
Education Method	Staff lecturers, practical cases, laboratory work, self-study. Depending on the health situation with regard to Covid-19 (part of) the lectures might be online.	
Assessment	Evaluation of individual oral presentations and intermediate and final written assignments. Details will be presented in the syllabus for the minor.	

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